



Top 54 CAT Logarithms Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

If $\log_3 2, \log_3(2^x - 5), \log_3(2^x - 7/2)$ are in arithmetic progression, then the value of x is equal to

- A 5
- B 4
- C 2
- D 3

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If $\log_3 2, \log_3(2^x - 5), \log_3(2^x - 7/2)$ are in arithmetic progression, then the value of x is equal to

A) 5
B) 4
C) 2
D) 3

A, B, C
 $2^x = Y$

$\log_3 2 + \log_3(2^x - 7/2) = 2 \log_3(2^x - 5)$
 $\log_3 2 + \log_3(2^x - 7/2) = 2 \log_3(2^x - 5)$
 $\log_3 2 + \log_3(2^x - 7/2) = 2 \log_3(2^x - 5)$

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Question 2

Let $u = (\log_2 x)^2 - 6\log_2 x + 12$ where x is a real number. Then the equation $x^u = 256$, has

- A no solution for x
- B exactly one solution for x
- C exactly two distinct solutions for x
- D exactly three distinct solutions for x

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Let $u = (\log_2 x)^2 - 6\log_2 x + 12$ where x is a real number. Then the equation $x^u = 256$, has

A) no solution for x
B) exactly one solution for x
C) exactly two distinct solutions for x
D) exactly three distinct solutions for x

$\log_2 x = A$
 $u = A^2 - 6A + 12$
 $x^u = 256$
 $u = \log_x 256$
 $8 \log_x 2$
 $\frac{8}{A}$

 VIDEO SOLUTION

Question 3

If $x = -0.5$, then which of the following has the smallest value?

- A $2^{1/x}$
- B $1/x$
- C $1/x^2$
- D 2^x
- E $1/\sqrt{-x}$



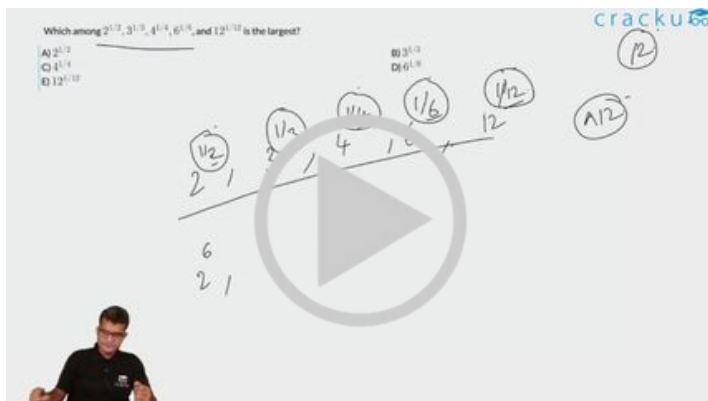
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Question 4

Which among $2^{1/2}$, $3^{1/3}$, $4^{1/4}$, $6^{1/6}$, and $12^{1/12}$ is the largest?

- A $2^{1/2}$
- B $3^{1/3}$
- C $4^{1/4}$
- D $6^{1/6}$
- E $12^{1/12}$



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Question 5

What are the values of x and y that satisfy both the equations?

$$2^{0.7x} * 3^{-1.25y} = 8\sqrt{6}/27$$

$$4^{0.3x} * 9^{0.2y} = 8 * 81^{1/5}$$

- A $x = 2, y = 5$
- B $x = 2.5, y = 6$
- C $x = 3, y = 5$
- D $x = 3, y = 4$
- E $x = 5, y = 2$

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Question 6

If $\log_y x = (a * \log_z y) = (b * \log_x z) = ab$, then which of the following pairs of values for (a, b) is not possible?

- A $(-2, 1/2)$
- B $(1, 1)$
- C $(0.4, 2.5)$
- D $(\pi, 1/\pi)$
- E $(2, 2)$

If $\log_y x = (a * \log_y y) = (b * \log_x z) = ab$, then which of the following pairs of values for (a, b) is not possible?

A) (-2, 1/2) B) (1, 1)
C) (0.4, 2.5) D) (π , $1/\pi$)
E) (2, 2)

Handwritten notes: $\log_y x = a * \log_y y = b * \log_x z = ab$, $\log_z y = b$, $y^z = z^b$, $y^z = x^{ab}$, $x^z = z^{ab}$.

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Question 7

If $x \geq y$ and $y > 1$, then the value of the expression $\log_x(x/y) + \log_y(y/x)$ can never be

- A -1
- B -0.5
- C 0
- D 1

If $x \geq y$ and $y > 1$, then the value of the expression $\log_x(x/y) + \log_y(y/x)$ can never be

A) -1 B) -0.5
C) 0 D) 1

Handwritten notes: $E = \log_x(x/y) + \log_y(y/x) = \log_x x - \log_x y + \log_y y - \log_y x = 1 - \log_x y + 1 - \log_y x = 2 - (\log_x y + \log_y x)$.

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Question 8

If $f(x) = \log \frac{(1+x)}{(1-x)}$, then $f(x) + f(y)$ is

- A $f(x+y)$
- B $f\left(\frac{x+y}{1+xy}\right)$
- C $(x+y)f\left(\frac{1}{1+xy}\right)$

D $\frac{f(x)+f(y)}{(1+xy)}$

[illegible]

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Question 9

$2^{73} - 2^{72} - 2^{71}$ is the same as

- A** 2^{69}
- B** 2^{70}
- C** 2^{71}
- D** 2^{72}

$2^{73} - 2^{72} - 2^{71}$ is the same as

A) 2^{69}
B) 2^{70}
C) 2^{71}
D) 2^{72}

Handwritten notes on the screen:

$2^{73} - 2^{72} - 2^{71}$ is the same as

$2^{71}(2^2 - 2^1 - 1)$

$2^{71}(4 - 2 - 1)$

$2^{71}(1)$

2^{71}

Handwritten calculations for the answer:

$2^{73} - 2^{72} - 2^{71}$

$2^{71}(2^2 - 2^1 - 1)$

$2^{71}(4 - 2 - 1)$

$2^{71}(1)$

2^{71}

Handwritten calculations for the answer:

$2^{73} - 2^{72} - 2^{71}$

$2^{71}(2^2 - 2^1 - 1)$

$2^{71}(4 - 2 - 1)$

$2^{71}(1)$

2^{71}

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CAT Previous Papers with detailed video solutions and analysis

Question 10

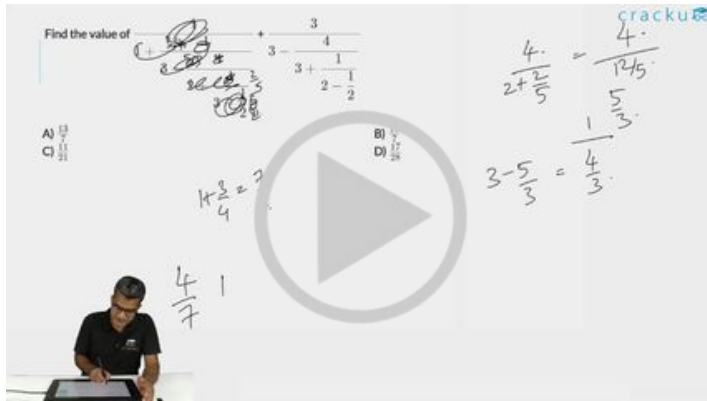
Find the value of $\frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{1}{3 - \frac{1}{2}}}}} + \frac{3}{3 - \frac{4}{3 + \frac{1}{2 - \frac{1}{2}}}}$

A $\frac{13}{7}$

B $\frac{15}{7}$

C $\frac{11}{21}$

D $\frac{17}{28}$



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Question 11

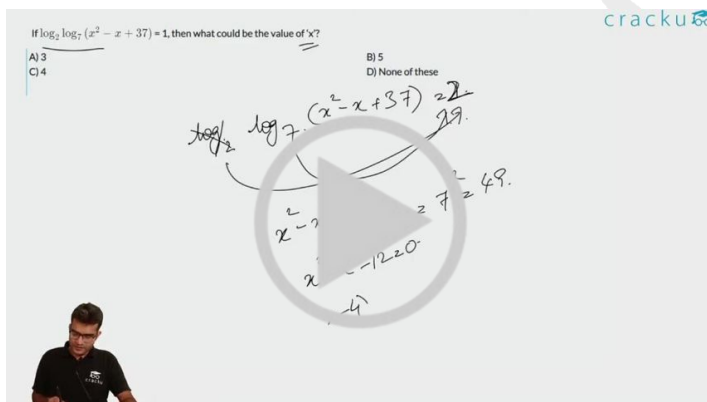
If $\log_2 \log_7 (x^2 - x + 37) = 1$, then what could be the value of 'x'?

A 3

B 5

C 4

D None of these



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Question 12

Which of the following is true?

A $7^{(3^2)} = (7^3)^2$

B $7^{(3^2)} > (7^3)^2$

C $7^{(3^2)} < (7^3)^2$

D None of these

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Question 13

Suppose, $\log_3 x = \log_{12} y = a$, where x, y are positive numbers. If G is the geometric mean of x and y , and $\log_6 G$ is equal to

A $\sqrt{a}$

B $2a$

C $a/2$

D a

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Suppose, $\log_3 x = \log_{12} y = a$, where x, y are positive numbers. If G is the geometric mean of x and y , and $\log_6 G$ is equal to

$x = 3^a$
 $y = 12^a$
 $G = \sqrt{3^a \cdot 12^a}$
 $\log_6 G = \frac{a}{2}$

[VIDEO SOLUTION](#)

Question 14

The value of $\log_{0.008} \sqrt{5} + \log_{\sqrt{3}} 81 - 7$ is equal to

A $1/3$

B $2/3$

C $5/6$

D $7/6$

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- A** 3
- B** $9/4$
- C** $4/9$
- D** $1/3$

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- A** $0 < x < 3$
- B** $23 < x < 30$
- C** $x > 30$

D $3 < x < 23$

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If x is a real number such that $\log_3 5 = \log_5 (2 + x)$, then which of the following is true?

A) $0 < x < 3$
B) $23 < x < 30$
C) $x > 30$
D) $3 < x < 23$

$\log_3 5 > \log_3 3 = 1$
 $\log_3 5 - \log_3 3 = 2$



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Question 17

If $9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$, then x is

- A $3/2$
- B $2/5$
- C $3/4$
- D $4/9$

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If $9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$, then x is

$9^{x-\frac{1}{2}} = 2^{2x-2}$
 $3^{2x-1} = 2^{2x-2}$
 $3^{2x-1} = 2^{2x-2}$
 $3^{2x-1} = 2^{2x-2}$

$3^{2x} = a$
 $3^{2x-3} = \frac{a}{3^3} = \frac{a}{27}$
 $\frac{a}{3} + \frac{a}{27}$



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Question 18

If $\log(2^a \times 3^b \times 5^c)$ is the arithmetic mean of $\log(2^2 \times 3^3 \times 5)$, $\log(2^6 \times 3 \times 5^7)$, and $\log(2 \times 3^2 \times 5^4)$, then a equals

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Question 19

A $\log_5 8$

B $1 + \log_3(\frac{5}{3})$

C $\log_5 9$

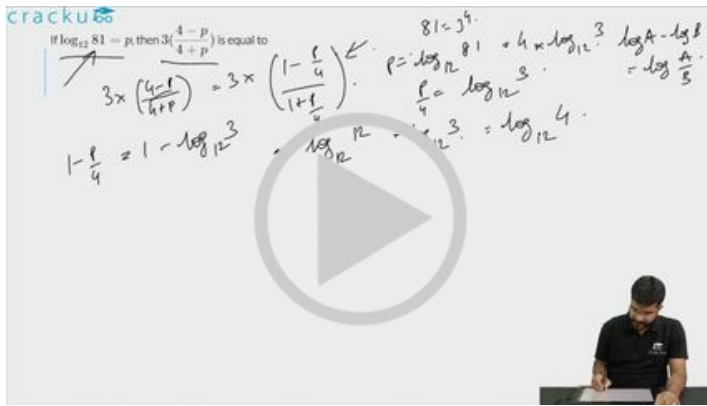
D $1 + \log_5(\frac{3}{5})$

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If $\log_{12} 81 = p$, then $3\left(\frac{4-p}{4+p}\right)$ is equal to

- A** $\log_4 16$
- B** $\log_6 16$
- C** $\log_2 8$

D $\log_6 8$

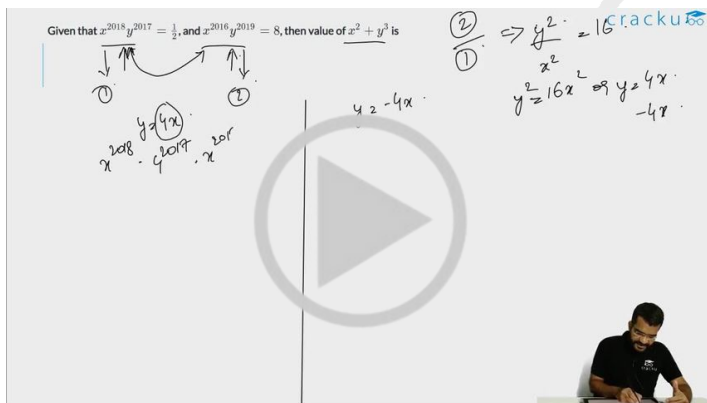


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Question 21

Given that $x^{2018}y^{2017} = \frac{1}{2}$, and $x^{2016}y^{2019} = 8$, then value of $x^2 + y^3$ is

- A $\frac{31}{4}$
- B $\frac{35}{4}$
- C $\frac{37}{4}$
- D $\frac{33}{4}$



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CAT Percentile Predictor



Question 22

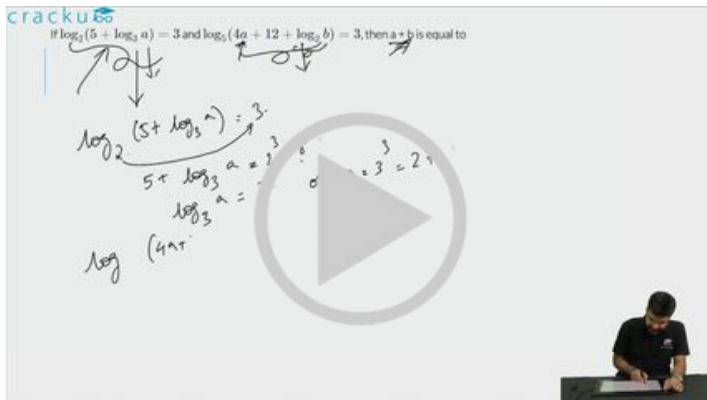
If $\log_2(5 + \log_3 a) = 3$ and $\log_5(4a + 12 + \log_2 b) = 3$, then $a + b$ is equal to

- A 59

B 40

C 32

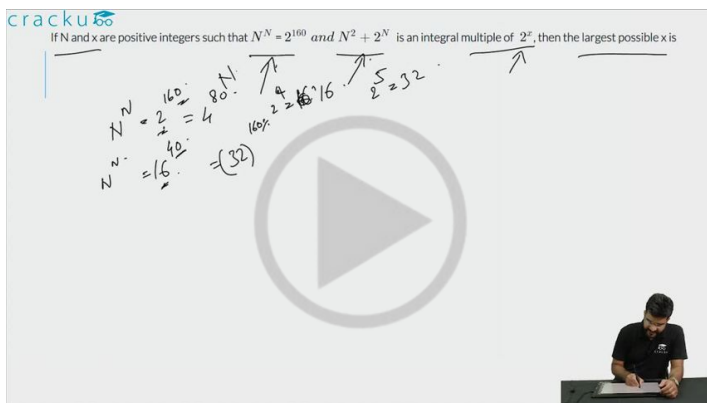
D 67



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Question 23

If N and x are positive integers such that $N^N = 2^{160}$ and $N^2 + 2^N$ is an integral multiple of 2^x , then the largest possible x is



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Question 24

$$\frac{1}{\log_2 100} - \frac{1}{\log_4 100} + \frac{1}{\log_5 100} - \frac{1}{\log_{10} 100} + \frac{1}{\log_{20} 100} - \frac{1}{\log_{25} 100} + \frac{1}{\log_{50} 100} = ?$$

A $\frac{1}{2}$

B 10

C 0

D -4

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$\frac{1}{\log_2 100} - \frac{1}{\log_3 100} + \frac{1}{\log_4 100} - \frac{1}{\log_5 100} + \frac{1}{\log_6 100} - \frac{1}{\log_7 100} + \frac{1}{\log_8 100} = ?$

$\log_2 2 - \log_{100} 4 + \log_{100} 5 - \log_{100} 10$
 $+ \log_{100} 20 - \log_{100} 10$

$\log_2 A + \log_2 C = \log_2 (A \times C)$
 $\log_2 A = \frac{1}{\log_2 B}$

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CAT Score Calculator

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Question 25

If $p^3 = q^4 = r^5 = s^6$, then the value of $\log_s(pqr)$ is equal to

- A $\frac{47}{10}$
- B $\frac{24}{5}$
- C $\frac{16}{5}$
- D 1

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If $p^3 = q^4 = r^5 = s^6$, then the value of $\log_s(pqr)$ is equal to

$p^3 = q^4 = r^5 = s^6 = k$
 $p = k^{1/3}$
 $q = k^{1/4}$
 $r = k^{1/5}$
 $s = k^{1/6}$

$pqr = k^{1/3} \times k^{1/4} \times k^{1/5}$
 $= k^{20+15+12/60} = k^{47/60}$

$\log_s(pqr) = \frac{\log(pqr)}{\log s} = \frac{47/60}{1/6} = \frac{47}{10}$

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Question 26

The real root of the equation $2^{6x} + 2^{3x+2} - 21 = 0$ is

- A $\log_2 9$
- B $\frac{\log_2 3}{3}$
- C $\log_2 27$

D $\frac{\log_2 7}{3}$

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The real root of the equation $2^{6x} + 2^{3x+2} - 21 = 0$ is

A) $\log_2 9$
B) $\log_2 3$
C) $\log_2 27$
D) $\frac{\log_2 7}{3}$

Handwritten solution:
 $2^{6x} + 2^{3x+2} - 21 = 0$
 $2^{6x} + 2^{3x} \cdot 2^2 - 21 = 0$
 $2^{6x} + 4 \cdot 2^{3x} - 21 = 0$
 $2^{6x} + 4 \cdot 2^{3x} - 21 = 0$
 $2^{6x} + 4 \cdot 2^{3x} - 21 = 0$
 $2^{6x} + 4 \cdot 2^{3x} - 21 = 0$

Video player interface showing a man writing on a board.

VIDEO SOLUTION

Question 27

If $(5.55)^x = (0.555)^y = 1000$, then the value of $\frac{1}{x} - \frac{1}{y}$ is

- A $\frac{1}{3}$
B 3
C 1
D $\frac{2}{3}$

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If $(5.55)^x = (0.555)^y = 1000$, then the value of $\frac{1}{x} - \frac{1}{y}$ is

Handwritten solution:
 $5.55^x = 1000$
 $x = \log_{5.55} 1000$
 $\frac{1}{x} = \log_{1000} 5.55$
 $\frac{1}{x} - \frac{1}{y} = \log_{1000} 5.55 - \log_{1000} 0.555$
 $\frac{1}{x} - \frac{1}{y} = \log_{1000} \frac{5.55}{0.555}$
 $\frac{1}{x} - \frac{1}{y} = \log_{1000} 10$
 $\frac{1}{x} - \frac{1}{y} = \frac{1}{3}$

Video player interface showing a man writing on a board.

VIDEO SOLUTION

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Question 28

If m and n are integers such that $(\sqrt{2})^{19} 3^4 4^2 9^m 8^n = 3^n 16^m (\sqrt[4]{64})$ then m is

- A -20
B -24

C -12

D -16

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If m and n are integers such that $(\sqrt{2})^{10} 3^4 4^2 9^m 8^n = 3^n 16^m (\sqrt[4]{64})^{4+2m}$ then m is

LHS = $2^{9.5} \times 3^4 \times 2^4 \times \frac{3^{2m}}{2^{2m}} \times 2^{3n} = 2^{9.5+4+3n-2m} \times 3^{4+2m}$

RHS = $3^n \times 2^{4m} \times 2^{4+2m} \times 3$



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Question 29

Let x and y be positive real numbers such that

$\log_5(x+y) + \log_5(x-y) = 3$, and $\log_2 y - \log_2 x = 1 - \log_2 3$. Then xy equals

A 150

B 25

C 100

D 250

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Let x and y be positive real numbers such that $\log_5(x+y) + \log_5(x-y) = 3$, and $\log_2 y - \log_2 x = 1 - \log_2 3$. Then xy equals

$\log_5(x+y)(x-y) = 3 \Rightarrow \log_5(x^2 - y^2) = 3 \Rightarrow x^2 - y^2 = 5^3 = 125 \quad \text{--- (1)}$

$\log_2 y - \log_2 x = 1 - \log_2 3 \Rightarrow \log_2 \frac{y}{x} = 1 - \log_2 3 \Rightarrow \frac{y}{x} = 2^{1 - \log_2 3} = \frac{2}{3} \Rightarrow y = \frac{2x}{3}$

Substituting $y = \frac{2x}{3}$ in (1):

$x^2 - \left(\frac{2x}{3}\right)^2 = 125 \Rightarrow x^2 - \frac{4x^2}{9} = 125 \Rightarrow \frac{5x^2}{9} = 125 \Rightarrow x^2 = 225 \Rightarrow x = 15$

$y = \frac{2x}{3} = \frac{2 \times 15}{3} = 10$

$xy = 15 \times 10 = 150$



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Question 30

If Y is a negative number such that $2^{Y^2(\log_3 5)} = 5^{\log_2 3}$, then Y equals to:

A $\log_2\left(\frac{1}{5}\right)$

B $\log_2\left(\frac{1}{3}\right)$

C $-\log_2\left(\frac{1}{5}\right)$

D $-\log_2\left(\frac{1}{3}\right)$

If Y is a negative number such that $2^{Y^2(\log_2 5)} = 5^{\log_2 3}$, then Y equals to:

A) $\log_2\left(\frac{1}{3}\right)$ B) $\log_2\left(\frac{1}{5}\right)$
 C) $-\log_2\left(\frac{1}{3}\right)$ D) $-\log_2\left(\frac{1}{5}\right)$

Handwritten solution: $Y^2 = \frac{\log_2 3}{\log_2 5} \times \log_2 5$

Handwritten solution: $2^{Y^2} = 5^{B/A}$
 $2^{Y^2} = 5^{B/A}$
 $Y^2 = \log_2 5 \times \log_2 5$
 $Y = \frac{B}{A}$

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Question 31

If $x = (4096)^{7+4\sqrt{3}}$, then which of the following equals to 64?

- A $\frac{x^7}{x^{2\sqrt{3}}}$
- B $\frac{x^{\frac{7}{2}}}{x^{\frac{4}{\sqrt{3}}}}$
- C $\frac{x^{\frac{7}{2}}}{x^{2\sqrt{3}}}$
- D $\frac{x^7}{x^{4\sqrt{3}}}$

Handwritten solution: $64^2 = 4096$

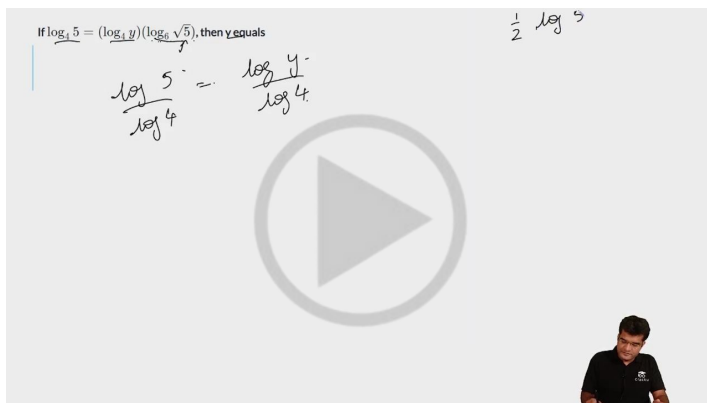
Handwritten solution: $x = (4096)^{7+4\sqrt{3}}$
 $4096 = x^{\frac{1}{7+4\sqrt{3}}}$

Handwritten solution: $\frac{1}{(7+4\sqrt{3})} \times \frac{7-4\sqrt{3}}{7-4\sqrt{3}}$
 $= \frac{7-4\sqrt{3}}{7^2 - (4\sqrt{3})^2}$

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Question 32

If $\log_4 5 = (\log_4 y)(\log_6 \sqrt{5})$, then y equals



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Question 33

If $\log_a 30 = A$, $\log_a \left(\frac{5}{3}\right) = -B$ and $\log_2 a = \frac{1}{3}$, then $\log_3 a$ equals

- A $\frac{2}{A+B-3}$
- B $\frac{2}{A+B} - 3$
- C $\frac{A+B}{2} - 3$
- D $\frac{A+B-3}{2}$



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Question 34

If a, b, c are non-zero and $14^a = 36^b = 84^c$, then $6b\left(\frac{1}{c} - \frac{1}{a}\right)$ is equal to

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If a, b, c are non-zero and $14^a = 36^b = 84^c$, then $6b\left(\frac{1}{c} - \frac{1}{a}\right)$ is equal to

$14^a = 36^b = 84^c = K$

$a = \log_{14} K$

$b = \log_{36} K$

$c = \log_{84} K$

$\frac{1}{c} = \log_K 84$

$\frac{1}{a} = \log_K 14$

$\frac{1}{c} - \frac{1}{a} = \log_K 84 - \log_K 14$

$6 \times b \times \left(\frac{1}{c} - \frac{1}{a}\right)$

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Question 35

$\frac{2 \times 4 \times 8 \times 16}{(\log_2 4)^2 (\log_4 8)^3 (\log_8 16)^4}$ equals

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$\frac{2 \times 4 \times 8 \times 16}{(\log_2 4)^2 (\log_4 8)^3 (\log_8 16)^4}$ equals

Denominator = $2^2 \times$

$\log_4 8 = \left(\frac{3}{2}\right) \times \log_2 2$

$\log_x x^m = m \times \log_x x$

$\log_y x^m = \frac{1}{m} \times \log_y x$

VIDEO SOLUTION

Question 36

The value of $\log_a\left(\frac{a}{b}\right) + \log_b\left(\frac{b}{a}\right)$, for $1 < a \leq b$ cannot be equal to

- A 0
- B -1
- C 1
- D -0.5

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The value of $\log_a\left(\frac{b}{a}\right) + \log_b\left(\frac{a}{b}\right)$, for $1 < a \leq b$ cannot be equal to

A) 0
B) -1
C) 1
D) -0.5

$\log_a \frac{b}{a} + \log_b \frac{a}{b}$
 $\log_a b - \log_a a + \log_b a - \log_b b$
 $\log_a b - 1 + \log_b a - 1$
 $\log_a b + \log_b a - 2$
 $1 - 1$
 0

$\log_a b = \gamma$
 $\log_b a = 1/\gamma$



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Question 37

If $5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = \log_{10} \frac{1}{\sqrt{1-x^2}}$, then $100x$ equals

If $5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = \log_{10} \frac{1}{\sqrt{1-x^2}}$, then $100x$ equals

Answer: _____

5 - 1

$1+x > 0 \Rightarrow -1 < x \leq 1$
 $1-x > 0 \Rightarrow x < 1$
 $\log \frac{1}{\sqrt{1-x^2}} = -\log \sqrt{1-x^2}$
 $= -(\log \sqrt{1-x} + \log \sqrt{1+x})$



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Question 38

If $\log_2[3 + \log_3\{4 + \log_4(x-1)\}] - 2 = 0$ then $4x$ equals

If $\log_2[3 + \log_3\{4 + \log_4(x-1)\}] - 2 = 0$ then $4x$ equals

$\log_2[3 + \log_3\{4 + \log_4(x-1)\}] = 2$

$3 + \log_3\{4 + \log_4(x-1)\} = 4$

$\log_3\{4 + \log_4(x-1)\} = 1$

$4 + \log_4(x-1) = 3$

$\log_4(x-1) = -1$

$x-1 = 4^{-1}$

$x = \frac{5}{4}$

$4x = 5$



Question 39

For a real number a , if $\frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$ then a must lie in the range

- A) $2 < a < 3$
- B) $3 < a < 4$
- C) $4 < a < 5$
- D) $a > 5$

For a real number a , if $\frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$ then a must lie in the range

A) $2 < a < 3$ B) $3 < a < 4$
 C) $4 < a < 5$ D) $a > 5$

Handwritten solution steps:

$$\frac{\log_{15} a + \log_{32} a}{\log_{15} a \cdot \log_{32} a} = 4$$

$$\frac{1}{\log_{15} a} + \frac{1}{\log_{32} a} = 4$$

$$\frac{1}{\log_{15} a} = 4 - \frac{1}{\log_{32} a}$$

$$\frac{1}{\log_{15} a} = \frac{4 \log_{32} a - 1}{\log_{32} a}$$

$$\log_{15} a = \frac{\log_{32} a}{4 \log_{32} a - 1}$$

$$\log_{15} a = \frac{\log_{32} a}{4 \log_{32} a - 1}$$

$$\log_{15} a = \frac{\log_{32} a}{4 \log_{32} a - 1}$$

CAT Study Material
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Question 40

If $(\sqrt{\frac{7}{5}})^{3x-y} = \frac{875}{2401}$ and $(\frac{4a}{b})^{6x-y} = (\frac{2a}{b})^{y-6x}$, for all non-zero real values of a and b , then the value of $x + y$ is

If $(\sqrt{\frac{7}{5}})^{3x-y} = \frac{875}{2401}$ and $(\frac{4a}{b})^{6x-y} = (\frac{2a}{b})^{y-6x}$, for all non-zero real values of a and b , then the value of $x + y$ is

Answer: $\frac{11}{5}$

Handwritten solution steps:

$$\left(\sqrt{\frac{7}{5}}\right)^{3x-y} = \frac{875}{2401}$$

$$\left(\frac{7}{5}\right)^{\frac{3x-y}{2}} = \frac{875}{2401}$$

$$\left(\frac{7}{5}\right)^{\frac{3x-y}{2}} = \left(\frac{7}{5}\right)^3$$

$$\frac{3x-y}{2} = 3$$

$$3x-y = 6$$

$$y = 3x - 6$$

$$\left(\frac{4a}{b}\right)^{6x-y} = \left(\frac{2a}{b}\right)^{y-6x}$$

$$\left(\frac{4a}{b}\right)^{6x-(3x-6)} = \left(\frac{2a}{b}\right)^{(3x-6)-6x}$$

$$\left(\frac{4a}{b}\right)^{3x+6} = \left(\frac{2a}{b}\right)^{-3x+6}$$

$$\left(\frac{4a}{b}\right)^{3x+6} = \left(\frac{2a}{b}\right)^{6-3x}$$

$$\left(\frac{4a}{b}\right)^{3x+6} = \left(\frac{2a}{b}\right)^{6-3x}$$

Question 41

If x and y are positive real numbers such that $\log_x(x^2 + 12) = 4$ and $3 \log_y x = 1$, then $x + y$ equals

- A 20
- B 68
- C 10
- D 11

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If x and y are positive real numbers such that $\log_x(x^2 + 12) = 4$ and $3 \log_y x = 1$, then $x + y$ equals

A) 20
B) 68
C) 10
D) 11

$x^2 = A$
 $x^4 = A^2$

$x^2 + 12 = A^2$
 $A^2 - A^2 = 0$
 $(A^2 - A^2) = 0$
 $A = 1$ or $A = 11$

VIDEO SOLUTION

Question 42

If $\sqrt{5x + 9} + \sqrt{5x - 9} = 3(2 + \sqrt{2})$, then $\sqrt{10x + 9}$ is equal to

- A $3\sqrt{31}$
- B $4\sqrt{5}$
- C $3\sqrt{7}$
- D $2\sqrt{7}$

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If $\sqrt{5x + 9} + \sqrt{5x - 9} = 3(2 + \sqrt{2})$, then $\sqrt{10x + 9}$ is equal to

A) $3\sqrt{31}$
B) $4\sqrt{5}$
C) $3\sqrt{7}$
D) $2\sqrt{7}$

$\sqrt{5x + 9} = k - \sqrt{5x - 9}$
 $5x + 9 = k^2 - 2\sqrt{5x - 9} + 5x - 9$
 $2\sqrt{5x - 9} = k^2 - 18$

VIDEO SOLUTION

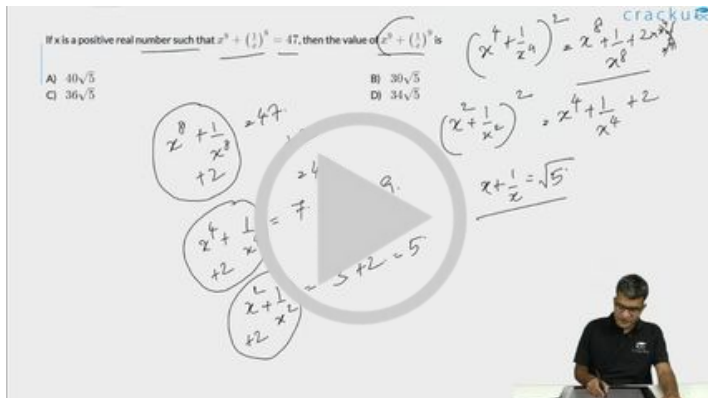
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CAT Daily Articles

Question 43

If x is a positive real number such that $x^8 + \left(\frac{1}{x}\right)^8 = 47$, then the value of $x^9 + \left(\frac{1}{x}\right)^9$ is

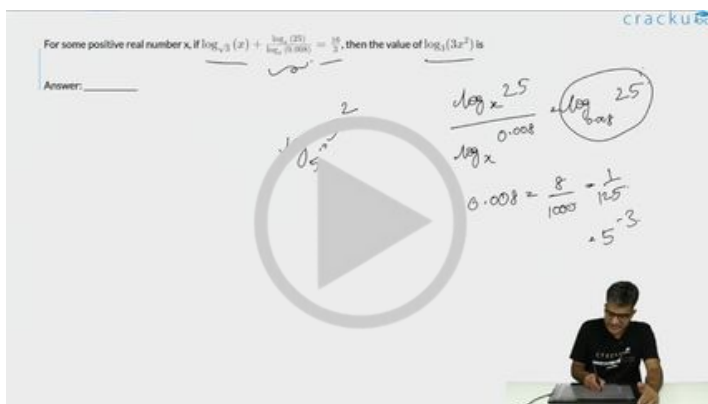
- A $40\sqrt{5}$
- B $30\sqrt{5}$
- C $36\sqrt{5}$
- D $34\sqrt{5}$



VIDEO SOLUTION

Question 44

For some positive real number x , if $\log_{\sqrt{3}}(x) + \frac{\log_x(25)}{\log_x(0.008)} = \frac{16}{3}$, then the value of $\log_3(3x^2)$ is



VIDEO SOLUTION

Question 45

If a , b and c are positive real numbers such that $a > 10 \geq b \geq c$ and $\frac{\log_8(a+b)}{\log_2 c} + \frac{\log_{27}(a-b)}{\log_3 c} = \frac{2}{3}$, then the greatest possible integer value of a is

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If a, b and c are positive real numbers such that $a > 10 \geq b \geq c$ and $\frac{\log_8(a+b)}{\log_2 c} + \frac{\log_2(a-b)}{\log_3 c} = \frac{2}{3}$, then the greatest possible integer value of a is

Answer: _____

The handwritten solution shows the following steps:

$$\log_8 \frac{a+b}{c} + \frac{\log_2(a-b)}{\log_3 c} = \frac{2}{3}$$

$$\frac{\log_2(a+b)}{3 \log_2 c} + \frac{\log_2(a-b)}{\log_3 c} = \frac{2}{3}$$

$$\frac{\log_2(a+b)}{3 \log_2 c} + \frac{\log_2(a-b)}{\log_2 c} = \frac{2}{3}$$

$$\frac{\log_2(a+b) + 3 \log_2(a-b)}{3 \log_2 c} = \frac{2}{3}$$

$$\log_2(a+b) + 3 \log_2(a-b) = 2 \log_2 c$$

$$\log_2(a+b) + \log_2(a-b)^3 = \log_2 c^2$$

$$\log_2(a+b)(a-b)^3 = \log_2 c^2$$

$$(a+b)(a-b)^3 = c^2$$

The video player shows a man writing the solution on a tablet.

VIDEO SOLUTION

100+ CAT Quant Questions



Question 46

If $(x + 6\sqrt{2})^{\frac{1}{2}} - (x - 6\sqrt{2})^{\frac{1}{2}} = 2\sqrt{2}$, then x equals

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If $(x + 6\sqrt{2})^{\frac{1}{2}} - (x - 6\sqrt{2})^{\frac{1}{2}} = 2\sqrt{2}$, then x equals

Answer: _____

The handwritten solution shows the following steps:

$$(x + 6\sqrt{2})^{\frac{1}{2}} - (x - 6\sqrt{2})^{\frac{1}{2}} = 2\sqrt{2}$$

$$(x + 6\sqrt{2})^{\frac{1}{2}} = (x - 6\sqrt{2})^{\frac{1}{2}} + 2\sqrt{2}$$

$$(x + 6\sqrt{2}) = (x - 6\sqrt{2}) + 4\sqrt{2}(x - 6\sqrt{2})^{\frac{1}{2}} + 8$$

$$x + 6\sqrt{2} = x - 6\sqrt{2} + 4\sqrt{2}(x - 6\sqrt{2})^{\frac{1}{2}} + 8$$

$$12\sqrt{2} - 8 = 4\sqrt{2}(x - 6\sqrt{2})^{\frac{1}{2}}$$

$$3\sqrt{2} - 2 = \sqrt{2}(x - 6\sqrt{2})^{\frac{1}{2}}$$

$$3 - \sqrt{2} = (x - 6\sqrt{2})^{\frac{1}{2}}$$

$$(3 - \sqrt{2})^2 = x - 6\sqrt{2}$$

$$9 - 6\sqrt{2} + 2 = x - 6\sqrt{2}$$

$$11 = x$$

The video player shows a man writing the solution on a tablet.

VIDEO SOLUTION

Question 47

If x is a positive real number such that $4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13$, then the greatest integer not exceeding x , is

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If x is a positive real number such that $4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13$, then the greatest integer not exceeding x , is

Answer: _____

Handwritten notes:

$$\log_B A = \frac{\log A}{\log B}$$

$$\log_{10} x + \frac{4}{2} \log_{10} x + \frac{8}{3} \log_{10} x = 13$$

$$\log_{10} x + 2 \log_{10} x + \frac{8}{3} \log_{10} x = 13$$

$$3 \log_{10} x + 4 \log_{10} x + 8 \log_{10} x = 39$$

$$15 \log_{10} x = 39$$

$$\log_{10} x = \frac{13}{5}$$

$$x = 10^{\frac{13}{5}}$$

Video player interface with a play button.

VIDEO SOLUTION

Question 48

If $(a + b\sqrt{n})$ is the positive square root of $(29 - 12\sqrt{5})$, where a and b are integers, and n is a natural number, then the maximum possible value of $(a + b + n)$ is

- A 18
- B 22
- C 4
- D 6

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If $(a + b\sqrt{n})$ is the positive square root of $(29 - 12\sqrt{5})$, where a and b are integers, and n is a natural number, then the maximum possible value of $(a + b + n)$ is

Handwritten notes:

$$(a + b\sqrt{n})^2 = 29 - 12\sqrt{5}$$

$$a^2 + b^2n + 2ab\sqrt{n} = 29 - 12\sqrt{5}$$

$$a^2 + b^2n = 29$$

$$2ab\sqrt{n} = -12\sqrt{5}$$

$$ab\sqrt{n} = -6\sqrt{5}$$

$$a^2b^2n = 36 \times 5 = 180$$

$$\frac{a^2 + b^2n}{a^2} = \frac{29}{a^2}$$

$$\frac{a^2 + 180}{a^2} = \frac{29}{a^2}$$

$$a^2 + 180 = 29$$

$$a^2 = -151$$

Video player interface with a play button.

VIDEO SOLUTION

100+ CAT DILR Questions



Question 49

If $(a + b\sqrt{3})^2 = 52 + 30\sqrt{3}$, where a and b are natural numbers, then $a + b$ equals

- A 7
- B 8
- C 9

D 10

If $(a + b\sqrt{3})^2 = 52 + 30\sqrt{3}$, where a and b are natural numbers, then $a + b$ equals

A) 7
B) 8
C) 9
D) 10

Handwritten notes: $2ab = 30$, $ab = 15$, $1, 15$, $a + 3b^2 = 52 + 2\sqrt{3}$, $2 + 3b^2 = 52$

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VIDEO SOLUTION

Question 50

The sum of all distinct real values of x that satisfy the equation $10^x + \frac{4}{10^x} = \frac{81}{2}$, is

- A) $2 \log_{10} 2$
B) $4 \log_{10} 2$
C) $\log_{10} 2$
D) $3 \log_{10} 2$

The sum of all distinct real values of x that satisfy the equation $10^x + \frac{4}{10^x} = \frac{81}{2}$, is

A) $2 \log_{10} 2$
B) $4 \log_{10} 2$
C) $\log_{10} 2$
D) $3 \log_{10} 2$

Handwritten notes: $10^x = A$, $A + \frac{4}{A} = \frac{81}{2}$, $A^2 - \frac{81}{2}A + 4 = 0$, $A_1 \times A_2 = 4$, $A_1 + A_2 = \frac{81}{2}$

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VIDEO SOLUTION

Question 51

If $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$ and $8^f = 9$, then the value of the product $abcde f$ is

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If $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$ and $8^f = 9$, then the value of the product $abcdef$ is

Answer:

$a = \log_3 4$ $e = \log_7 8$
 $b = \log_4 5$ $f = \log_8 9$
 $c =$
 $d =$

log A

VIDEO SOLUTION

100+ CAT VARC Questions

Question 52

The number of distinct integers n for which $\log_{\frac{1}{4}}(n^2 - 7n + 11) > 0$, is

- A 2
- B infinite
- C 1
- D 0

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Q1

$n^2 - 7n + 11 < 1$
 $n^2 - n + 10 < 0$
 $(n-3)(n-5) < 0$
 $(2, 5)$ $n = 3, 4$

If $B > 1$, $A > 1$
 If $B < 1$, $A < 1$

VIDEO SOLUTION

Question 53

The sum of all possible real values of x for which $\log_{x-3}(x^2 - 9) = \log_{x-3}(x + 1) + 2$, is

- A -3
- B $\sqrt{33}$
- C 3

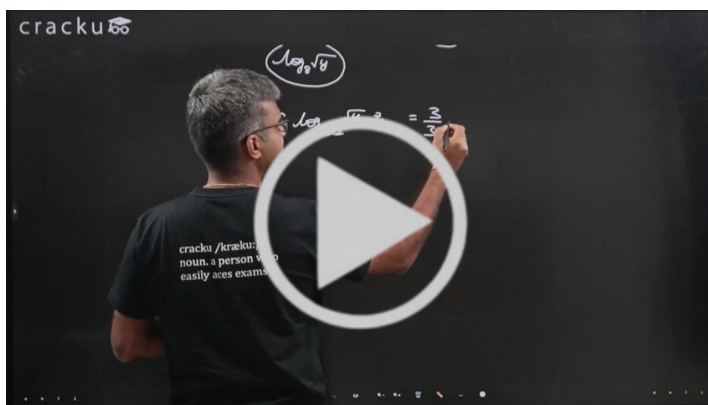
D $\frac{3 + \sqrt{33}}{2}$

[VIEW SOLUTION](#)

Question 54

If $\log_{64} x^2 + \log_8 \sqrt{y} + 3 \log_{512} (\sqrt{y}z) = 4$, where x,y and z are positive real numbers, then the minimum possible value of $(x + y + z)$ is

- A 48
- B 36
- C 24
- D 96



[VIDEO SOLUTION](#)

CAT Expected Questions PDF



Answers

1.D	2.B	3.B	4.B	5.E	6.E	7.D	8.B
9.C	10.B	11.C	12.B	13.D	14.C	15.B	16.D
17.A	18.3	19.D	20.D	21.D	22.A	23.10	24.A
25.A	26.B	27.A	28.C	29.A	30.B	31.C	32.36
33.A	34.3	35.24	36.C	37.99	38.5	39.C	40.14
41.C	42.C	43.D	44.7	45.14	46.11	47.31	48.A
49.B	50.A	51.2	52.D	53.D	54.A		

Explanations

1. **D**

$$2\log(2^x - 5) = \log 2 + \log(2^x - 7/2)$$

$$\text{Let } 2^x = t$$

$$\Rightarrow (t - 5)^2 = 2(t - 7/2)$$

$$\Rightarrow t^2 + 25 - 10t = 2t - 7$$

$$\Rightarrow t^2 - 12t + 32 = 0$$

$$\Rightarrow t = 8, 4$$

Therefore, $x = 2$ or 3 , but $2^x > 5$, so $x = 3$

 VIDEO SOLUTION

2. **B**

$$x^u = 256$$

Taking log to the base 2 on both the sides,

$$u * \log_2 x = \log_2 256$$

$$\Rightarrow [(\log_2 x)^2 - 6\log_2 x + 12] * \log_2 x = 8$$

$$(\log_2 x)^3 - 6(\log_2 x)^2 + 12\log_2 x = 8$$

$$\text{Let } \log_2 x = t$$

$$t^3 - 6t^2 + 12t - 8 = 0$$

$$(t - 2)^3 = 0$$

$$\text{Therefore, } \log_2 x = 2$$

$$\Rightarrow x = 4 \text{ is the only solution}$$

Hence, option B is the correct answer.

 VIDEO SOLUTION

3. **B**

2^p is always positive

x^2 is always non negative.

$1/\sqrt{-x}$ is always positive.

$\frac{1}{x}$ is negative when x is negative.

In this case, x is negative $\Rightarrow \frac{1}{x}$ is smallest.

 VIDEO SOLUTION

4. B

Make the power equal and compare the denominators.

$2^{1/2}$ can be written as $64^{1/12}$

$3^{1/3}$ can be written as $81^{1/12}$

$4^{1/4}$ can be written as $64^{1/12}$

$6^{1/6}$ can be written as $36^{1/12}$

Among these, $81^{1/12}$ is the greatest $\Rightarrow 3^{1/3}$ is the greatest.

 VIDEO SOLUTION

5. E

$$2^{0.7x} * 3^{-1.25y} = 8\sqrt{6}/27 \Rightarrow 2^{0.7x} * 3^{-1.25y} = 2^{3.5} * 3^{-2.5}$$

$$\Rightarrow 0.7x = 3.5 \Rightarrow x = 5$$

$$\Rightarrow -1.25y = -2.5 \Rightarrow y = 2$$

$$4^{0.3x} * 9^{0.2y} = 8 * 81^{1/5} \Rightarrow 2^{0.6x} * 3^{0.4y} = 2^3 * 3^{0.8}$$

$$\Rightarrow 0.6x = 3 \Rightarrow x = 5$$

$$\Rightarrow 0.4y = 0.8 \Rightarrow y = 2$$

$\Rightarrow (5, 2)$ is the solution.

 VIEW SOLUTION



CAT Important Topics

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6. E

$$\log_y x = ab$$

$$a * \log_z y = ab \Rightarrow \log_z y = b$$

$$b * \log_x z = ab \Rightarrow \log_x z = a$$

$$\log_y x = \log_z y * \log_x z \Rightarrow \log x / \log y = \log y / \log z * \log z / \log x$$

$$\Rightarrow \frac{\log x}{\log y} = \frac{\log y}{\log x}$$

$$\Rightarrow (\log x)^2 = (\log y)^2$$

$$\Rightarrow \log x = \log y \text{ or } \log x = -\log y$$

$$\text{So, } x = y \text{ or } x = 1/y$$

$$\text{So, } ab = 1 \text{ or } -1$$

Option 5) is not possible

 VIDEO SOLUTION

7. D

$$\log_x(x/y) + \log_y(y/x) = 1 - \log_x(y) + 1 - \log_y(x)$$

$$= 2 - (\log_x y + 1/\log_x y) \leq 0 \text{ (Since } \log_x y + 1/\log_x y \geq 2)$$

So, the value of the expression cannot be 1.

VIDEO SOLUTION

8. B

If $f(x) = \log \frac{(1+x)}{(1-x)}$ then $f(y) = \log \frac{(1+y)}{(1-y)}$

Also $\log (A*B) = \log A + \log B$

$$f(x)+f(y) = \log \frac{(1+x)(1+y)}{(1-x)(1-y)}$$

$$= \log \frac{(1+xy+x+y)}{(1+xy-x-y)}$$

Dividing numerator and denominator by $(1+xy)$

$$\log \frac{\frac{(1+xy+x+y)}{1+xy}}{\frac{(1+xy-x-y)}{1+xy}}$$

$$= \log \frac{\frac{1+xy}{1+xy} + \frac{(x+y)}{1+xy}}{\frac{1+xy}{1+xy} - \frac{(x+y)}{1+xy}}$$

$$= \log \frac{1 + \frac{(x+y)}{(1+xy)}}{1 - \frac{(x+y)}{(1+xy)}}$$

Hence option B.

VIDEO SOLUTION

9. C

$$2^{71}(2^2 - 2^1 - 1)$$

$$2^{71}(4 - 2 - 1)$$

$$2^{71}$$

VIDEO SOLUTION

10. B

$$1 + \frac{1}{3 - \frac{1}{2 + \frac{1}{3 - \frac{1}{2}}}}$$

$$= \frac{1}{1 + \frac{1}{3 - \frac{1}{2 + \frac{1}{3 - \frac{1}{2}}}}}$$

$$= \frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{2}{5}}}}$$

$$= \frac{1}{1 + \frac{1}{3 - \frac{20}{12}}}$$

$$= \frac{1}{1 + \frac{1}{3 - \frac{20}{12}}}$$

$$= \frac{1}{1 + \frac{12}{16}}$$

$$= \frac{1}{\frac{28}{16}}$$

$$= \frac{16}{28}$$

Similarly, Second term can be reduced to $\frac{33}{21}$ or $\frac{11}{7}$

Sum will be $= \frac{11}{7} + \frac{4}{7} = \frac{15}{7}$

VIDEO SOLUTION



11. C

$$\log_2 \log_7 (x^2 - x + 37) = 1$$

$$\log_7 (x^2 - x + 37) = 2$$

$$(x^2 - x + 37) = 7^2$$

Given eq. can be reduced to $x^2 - x + 37 = 49$

So x can be either -3 or 4.

[▶ VIDEO SOLUTION](#)

12. B

$$7^{(3^2)} = 7^9$$

$$(7^3)^2 = 7^6$$

$$\text{So } 7^{(3^2)} > (7^3)^2$$

[▶ VIEW SOLUTION](#)

13. D

We know that $\log_3 x = a$ and $\log_{12} y = a$

Hence, $x = 3^a$ and $y = 12^a$

Therefore, the geometric mean of x and y equals $\sqrt{x \times y}$

This equals $\sqrt{3^a \times 12^a} = 6^a$

Hence, $G = 6^a$ Or, $\log_6 G = a$

[▶ VIDEO SOLUTION](#)

14. C

$$\log_{0.008} \sqrt{5} + \log_{\sqrt{3}} 81 - 7$$

$$81 = 3^4 \text{ and } 0.008 = \frac{8}{1000} = \frac{2^3}{10^3} = \frac{1}{5^3} = 5^{-3}$$

Hence,

$$\log_{0.008} \sqrt{5} + 8 - 7$$

$$\log_{5^{-3}} 5^{\frac{1}{2}} + 8 - 7$$

$$\frac{\log 5^{0.5}}{\log 5^{-3}} + 1$$

$$-\frac{1}{6} + 1$$

$$= \frac{5}{6}$$

[▶ VIDEO SOLUTION](#)

15. B

$$\frac{81^x}{9} - \frac{81^x}{81} = 1944$$

$$81^x * \left[\frac{1}{9} - \frac{1}{81} \right] = 1944$$

$$81^x * \left[\frac{1}{81} \right] = 243$$

$$3^{4x} = 3^9$$

$$x = \frac{9}{4}$$

[VIDEO SOLUTION](#)



16. D

$$1 < \log_3 5 < 2$$

$$\Rightarrow 1 < \log_5(2+x) < 2$$

$$\Rightarrow 5 < 2+x < 25$$

$$\Rightarrow 3 < x < 23$$

[VIDEO SOLUTION](#)

17. A

$$\text{It is given that } 9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$$

Let us try to reduce them to powers of 3 and 2

$$\text{The given equation can be reduced to } 3^{2x-1} + 3^{2x-3} = 2^{2x} + 2^{2x-2}$$

$$\text{Hence, } 3^{2x-3} \times 10 = 2^{2x-2} \times 5$$

$$\text{Therefore, } 3^{2x-3} = 2^{2x-3}$$

$$\text{This is possible only if } 2x - 3 = 0 \text{ or } x = 3/2$$

[VIDEO SOLUTION](#)

18. 3

$$\log(2^a \times 3^b \times 5^c) = \frac{\log(2^2 \times 3^3 \times 5) + \log(2^6 \times 3 \times 5^7) + \log(2 \times 3^2 \times 5^4)}{3}$$

$$\log(2^a \times 3^b \times 5^c) = \frac{\log(2^{2+6+1} \times 3^{3+1+2} \times 5^{1+7+4})}{3}$$

$$\log(2^a \times 3^b \times 5^c) = \frac{\log(2^9 \times 3^6 \times 5^{12})}{3}$$

$$3\log(2^a \times 3^b \times 5^c) = \log(2^9 \times 3^6 \times 5^{12})$$

$$\text{Hence, } 3a = 9 \text{ or } a = 3$$

[VIDEO SOLUTION](#)

19. D

$$\text{Given that: } 2^x = 3^{\log_5 2}$$

$$\Rightarrow 2^x = 2^{\log_5 3}$$

$$\Rightarrow x = \log_5 3$$

$$\Rightarrow x = \log_5 \frac{3 \times 5}{5}$$

$$\Rightarrow x = \log_5 5 + \log_5 \frac{3}{5}$$

$$\Rightarrow x = 1 + \log_5 \frac{3}{5}. \text{ Hence, option D is the correct answer.}$$

VIDEO SOLUTION

20. D

Given that: $\log_{12} 81 = p$

$$\Rightarrow \log_{81} 12 = \frac{1}{p}$$

$$\Rightarrow \log_3 3 * 4 = \frac{4}{p}$$

$$\Rightarrow 1 + \log_3 4 = \frac{4}{p}$$

Using Componendo and Dividendo,

$$\Rightarrow \frac{1 + \log_3 4 - 1}{1 + \log_3 4 + 1} = \frac{4 - p}{4 + p}$$

$$\Rightarrow \frac{\log_3 4}{2 + \log_3 4} = \frac{4 - p}{4 + p}$$

$$\Rightarrow \frac{\log_3 4}{\log_3 9 + \log_3 4} = \frac{4 - p}{4 + p}$$

$$\Rightarrow \frac{\log_3 4}{\log_3 36} = \frac{4 - p}{4 + p}$$

$$\Rightarrow 3 * \frac{4 - p}{4 + p} = \frac{3 \log_3 4}{\log_3 36}$$

$$\Rightarrow 3 * \frac{4 - p}{4 + p} = \frac{\log_3 64}{\log_3 36}$$

$$\Rightarrow 3 * \frac{4 - p}{4 + p} = \log_{36} 64$$

$$\Rightarrow 3 * \frac{4 - p}{4 + p} = \log_{6^2} 8^2 = \log_6 8. \text{ Hence, option D is the correct answer.}$$

VIDEO SOLUTION



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21. D

Given that $x^{2018}y^{2017} = \frac{1}{2} \dots (1)$

$x^{2016}y^{2019} = 8 \dots (2)$

Equation (2)/ Equation (1)

$$\frac{y^2}{x^2} = \frac{8}{1/2}$$

$$\frac{y}{x} = 4 \text{ or } -4$$

Case 1: When $\frac{y}{x} = 4$

$$x^{2018}(4x)^{2017} = \frac{1}{2}$$

$$x^{2018+2017}(2)^{4034} = \frac{1}{2}$$

$$x^{4035} = \frac{1}{(2)^{4035}}$$

$$x = \frac{1}{2}$$

Since, $\frac{y}{x} = 4$, $\Rightarrow y = 2$

Therefore, $x^2 + y^3 = \frac{1}{4} + 8 = \frac{33}{4}$

Case 2: When $\frac{y}{x} = -4$

$$x^{2018}(-4x)^{2017} = \frac{1}{2}$$

$$x^{2018+2017}(2)^{4034} = \frac{-1}{2}$$

$$x^{4035} = \frac{1}{(-2)^{4035}}$$

$$x = \frac{-1}{2}$$

Since, $\frac{y}{x} = -4$, $\Rightarrow y = 2$

Therefore, $x^2 + y^3 = \frac{1}{4} + 8 = \frac{33}{4}$. Hence, option D is the correct answer.

 VIDEO SOLUTION

22. A

$$\log_2(5 + \log_3 a) = 3$$

$$\Rightarrow 5 + \log_3 a = 8$$

$$\Rightarrow \log_3 a = 3$$

$$\text{or } a = 27$$

$$\log_5(4a + 12 + \log_2 b) = 3$$

$$\Rightarrow 4a + 12 + \log_2 b = 125$$

Putting $a = 27$, we get

$$\log_2 b = 5$$

$$\text{or, } b = 32$$

$$\text{So, } a + b = 27 + 32 = 59$$

Hence, option A is the correct answer.

 VIDEO SOLUTION

23. 10

It is given that $N^N = 2^{160}$

We can rewrite the equation as $N^N = (2^5)^{160/5} = 32^{32}$

$$\Rightarrow N = 32$$

$$N^2 + 2^N = 32^2 + 2^{32} = 2^{10} + 2^{32} = 2^{10} * (1 + 2^{22})$$

Hence, we can say that $N^2 + 2^N$ can be divided by 2^{10}

Therefore, $x_{max} = 10$

▶ VIDEO SOLUTION

24. A

We know that $\frac{1}{\log_a b} = \frac{\log_x a}{\log_x b}$

Therefore, we can say that $\frac{1}{\log_2 100} = \frac{\log_{10} 2}{\log_{10} 100}$

$$\Rightarrow \frac{1}{\log_2 100} - \frac{1}{\log_4 100} + \frac{1}{\log_5 100} - \frac{1}{\log_{10} 100} + \frac{1}{\log_{20} 100} - \frac{1}{\log_{25} 100} + \frac{1}{\log_{50} 100}$$
$$\Rightarrow \frac{\log_{10} 2}{\log_{10} 100} - \frac{\log_{10} 4}{\log_{10} 100} + \frac{\log_{10} 5}{\log_{10} 100} - \frac{\log_{10} 10}{\log_{10} 100} + \frac{\log_{10} 20}{\log_{10} 100} - \frac{\log_{10} 25}{\log_{10} 100} + \frac{\log_{10} 50}{\log_{10} 100}$$

We know that $\log_{10} 100 = 2$

$$\Rightarrow \frac{1}{2} * [\log_{10} 2 - \log_{10} 4 + \log_{10} 5 - \log_{10} 10 + \log_{10} 20 - \log_{10} 25 + \log_{10} 50]$$
$$\Rightarrow \frac{1}{2} * [\log_{10} \frac{2 * 5 * 20 * 50}{4 * 10 * 25}]$$
$$\Rightarrow \frac{1}{2} * [\log_{10} 10]$$
$$\Rightarrow \frac{1}{2}$$

▶ VIDEO SOLUTION

25. A

Given that, $p^3 = q^4 = r^5 = s^6$

$$p^3 = s^6$$

$$p = s^{\frac{6}{3}} = s^2 \quad \dots(1)$$

$$\text{Similarly, } q = s^{\frac{6}{4}} = s^{\frac{3}{2}} \quad \dots(2)$$

$$\text{Similarly, } r = s^{\frac{6}{5}} \quad \dots(3)$$

$$\Rightarrow \log_s(pqr)$$

By substituting value of p, q, and r from equation (1), (2) and (3)

$$\Rightarrow \log_s(s^2 * s^{\frac{3}{2}} * s^{\frac{6}{5}})$$

$$\Rightarrow \log_s(s^{\frac{47}{10}})$$

$$\Rightarrow \frac{47}{10}$$

Hence, option A is the correct answer.

▶ VIDEO SOLUTION



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26. B

$$\text{Let } 2^{3x} = v$$

$$2^{6x} + 2^{3x+2} - 21 = 0$$

$$= v^2 + 4v - 21 = 0$$

$$=(v+7)(v-3)=0$$

$$v=3, -7$$

$$2^{3x} = 3 \text{ or } 2^{3x} = -7 \text{ (This can be negated)}$$

$$3x = \log_2 3$$

$$x = \log_2 3 / 3$$

 VIDEO SOLUTION

27. A

$$\text{We have, } (5.55)^x = (0.555)^y = 1000$$

Taking log in base 10 on both sides,

$$x(\log_{10} 555 - 2) = y(\log_{10} 555 - 3) = 3$$

$$\text{Then, } x(\log_{10} 555 - 2) = 3 \dots (1)$$

$$y(\log_{10} 555 - 3) = 3 \dots (2)$$

From (1) and (2)

$$\Rightarrow \log_{10} 555 = \frac{3}{x} + 2 = \frac{3}{y} + 3$$

$$\Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{3}$$

 VIDEO SOLUTION

28. C

$$\text{We have, } (\sqrt{2})^{19} 3^4 4^{2m} 9^{3n} = 3^n 16^m (\sqrt[4]{64})$$

Converting both sides in powers of 2 and 3, we get

$$2^{\frac{19}{2}} 3^4 2^4 3^{2m} 2^{3n} = 3^n 2^{4m} 2^{\frac{6}{4}}$$

$$\text{Comparing the power of 2 we get, } \frac{19}{2} + 4 + 3n = 4m + \frac{6}{4}$$

$$\Rightarrow 4m = 3n + 12 \dots (1)$$

$$\text{Comparing the power of 3 we get, } 4 + 2m = n$$

Substituting the value of n in (1), we get

$$4m = 3(4 + 2m) + 12$$

$$\Rightarrow m = -12$$

 VIDEO SOLUTION

29. A

$$\text{We have, } \log_5 (x + y) + \log_5 (x - y) = 3$$

$$\Rightarrow x^2 - y^2 = 125 \dots (1)$$

$$\log_2 y - \log_2 x = 1 - \log_2 3$$

$$\Rightarrow \frac{y}{x} = \frac{2}{3}$$

$$\Rightarrow 2x = 3y \Rightarrow x = \frac{3y}{2}$$

On substituting the value of x in 1, we get

$$\frac{5x^2}{4} = 125$$

$$\Rightarrow y=10, x=15$$

$$\text{Hence } xy=150$$

 VIDEO SOLUTION

30. B

$$2^{Y^2(\log_3 5)} = 5^{Y^2(\log_3 2)}$$

$$\text{Given, } 5^{Y^2(\log_3 2)} = 5^{(\log_2 3)}$$

$$\Rightarrow Y^2(\log_3 2) = (\log_2 3) \Rightarrow Y^2 = (\log_2 3)^2$$

$$\Rightarrow Y = (-\log_2 3) \text{ or } (\log_2 3)$$

$$\text{since } Y \text{ is a negative number, } Y = (-\log_2 3) = (\log_2 \frac{1}{3})$$

 VIDEO SOLUTION



CAT Syllabus PDF



31. C

$$x = 2^{12(7+4\sqrt{3})}$$

$$x^{\frac{7}{2}} = 2^{42(7+4\sqrt{3})}$$

$$x^{2\sqrt{3}} = 2^{24\sqrt{3}(7+4\sqrt{3})}$$

$$\frac{x^{\frac{7}{2}}}{x^{2\sqrt{3}}} = 2^{(7+4\sqrt{3})(42-24\sqrt{3})} = 2^{(7+4\sqrt{3})(7-4\sqrt{3})6} = 2^6$$

Hence C is correct answer.

 VIDEO SOLUTION

32. 36

$$\frac{\log 5}{2 \log 2} = \frac{\log y}{2 \log 2} \cdot \frac{\log 5}{2 \log 6}$$

$$\log 36 = \log y; \therefore y = 36$$

 VIDEO SOLUTION

33. A

$$\log_a 30 = A \text{ or } \log_a 5 + \log_a 2 + \log_a 3 = A \dots \dots \dots (1)$$

$$\log_a \left(\frac{5}{3}\right) = -B \text{ or } \log_a 3 - \log_a 5 = B \dots \dots \dots (2)$$

$$\text{and finally } \log_a 2 = 3$$

$$\text{Substituting this in (1) we get } \log_a 5 + \log_a 3 = A - 3$$

Now we have two equations in two variables (1) and (2). On solving we get

$$\log_a 3 = \frac{(A+B-3)}{2} \text{ or } \log_3 a = \frac{2}{A+B-3}$$

 VIDEO SOLUTION

34. 3

$$\text{Let } 14^a = 36^b = 84^c = k$$

$$\Rightarrow a = \log_{14} k, b = \log_{36} k, c = \log_{84} k$$

$$6b\left(\frac{1}{c} - \frac{1}{a}\right) = 6 \cdot \frac{1}{2} \log_6 k (\log_k 84 - \log_k 14) = 3$$

[VIDEO SOLUTION](#)

35.24

$$\begin{aligned} & \frac{2 \times 4 \times 8 \times 16}{(\log_2 4)^2 (\log_4 8)^3 (\log_8 16)^4} \\ &= \frac{2^1 \times 2^2 \times 2^3 \times 2^4}{(\log_2 2^2)^2 (\log_2 2^3)^3 (\log_2 2^4)^4} \\ &= \frac{2^{1+2+3+4}}{(2)^2 \left(\frac{3}{2}\right)^3 \left(\frac{4}{2}\right)^4} \\ &= \frac{2^{10}}{4 \left(\frac{3}{2}\right)^3 \left(\frac{4}{3}\right)^4} = 24 \end{aligned}$$

[VIDEO SOLUTION](#)



36.C

On expanding the expression we get $1 - \log_a b + 1 - \log_b a$

$$\text{or } 2 - \left(\log_a b + \frac{1}{\log_a b}\right)$$

Now applying the property of AM $\geq$ GM, we get that $\frac{\left(\log_a b + \frac{1}{\log_a b}\right)}{2} \geq 1$ or $\left(\log_a b + \frac{1}{\log_a b}\right) \geq 2$ Hence from here we can conclude that the expression will always be equal to 0 or less than 0. Hence any positive value is not possible. So 1 is not possible.

[VIDEO SOLUTION](#)

37.99

$$5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = \log_{10} \frac{1}{\sqrt{1-x^2}}$$

We can re-write the equation as: $5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = \log_{10} (\sqrt{1+x} \times \sqrt{1-x})^{-1}$

$$5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = (-1) \log_{10} (\sqrt{1+x}) + (-1) \log_{10} (\sqrt{1-x})$$

$$5 = -\log_{10} \sqrt{1+x} + \log_{10} \sqrt{1+x} - \log_{10} \sqrt{1-x} - 4 \log_{10} \sqrt{1-x}$$

$$5 = -5 \log_{10} \sqrt{1-x}$$

$$\sqrt{1-x} = \frac{1}{10}$$

$$\text{Squaring both sides: } (\sqrt{1-x})^2 = \frac{1}{100}$$

$$\therefore x = 1 - \frac{1}{100} = \frac{99}{100}$$

$$\text{Hence, } 100x = 100 \times \frac{99}{100} = 99$$

[VIDEO SOLUTION](#)

38.5

We have :

$$\log_2 \{3 + \log_3 \{4 + \log_4 (x-1)\}\} = 2$$

$$\text{we get } 3 + \log_3 \{4 + \log_4 (x-1)\} = 4$$

$$\text{we get } \log_3 (4 + \log_4 (x - 1)) = 1)$$

$$\text{we get } 4 + \log_4 (x - 1) = 3$$

$$\log_4 (x - 1) = -1$$

$$x - 1 = 4^{-1}$$

$$x = \frac{1}{4} + 1 = \frac{5}{4}$$

$$4x = 5$$

 VIDEO SOLUTION

39. C

$$\text{We have : } \frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$$

$$\text{We get } \frac{\left(\frac{\log a}{\log 15} + \frac{\log a}{\log 32}\right)}{\frac{\log a}{\log 15} \times \frac{\log a}{\log 32}} = 4$$

$$\text{we get } \log a (\log 32 + \log 15) = 4 (\log a)^2$$

$$\text{we get } (\log 32 + \log 15) = 4 \log a$$

$$= \log 480 = \log a^4$$

$$= a^4 = 480$$

so we can say a is between 4 and 5 .

 VIDEO SOLUTION

40. 14

$$\left(\sqrt{\frac{7}{5}}\right)^{3x-y} = \frac{875}{2401}$$

$$\left(\frac{7}{5}\right)^{\frac{(3x-y)}{2}} = \frac{125}{343}$$

$$\left(\frac{7}{5}\right)^{\frac{(3x-y)}{2}} = \left(\frac{7}{5}\right)^{-3}$$

$$3x - y = -6$$

$$\left(\frac{4a}{b}\right)^{6x-y} = \left(\frac{2a}{b}\right)^{y-6x}$$

Therefore, $y=6x$ as the bases are different so the power should be zero for the results to be equal.

$$3x - y = -6$$

$$\text{or, } 3x - 6x = -6$$

$$\text{or } x = 2$$

$$y = 6x = 12$$

$$x + y = 14$$

 VIDEO SOLUTION



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41. C

$$\text{Given, } \log_x (x^2 + 12) = 4$$

$$\Rightarrow x^2 + 12 = x^4$$

$$\Rightarrow x^4 - x^2 - 12 = 0$$

$$\Rightarrow x^4 - 4x^2 + 3x^2 - 12 = 0$$

$$\Rightarrow x^2 (x^2 - 4) + 3 (x^2 - 4) = 0$$

$$\Rightarrow (x^2 - 4) (x^2 + 3) = 0 \Rightarrow \text{since, } x \text{ is a positive real number (given)} \Rightarrow x = 2.$$

Now, Given $3 \log_y x = 1$

$$\Rightarrow \log_y x = \frac{1}{3}$$

$$\Rightarrow x = y^{\frac{1}{3}}$$

$$\Rightarrow y = x^3 \Rightarrow y = 8.$$

$$\Rightarrow x + y = 2 + 8 = 10.$$

 VIDEO SOLUTION

42. C

$$\text{Given, } \sqrt{5x+9} + \sqrt{5x-9} = 3(2 + \sqrt{2})$$

$$\Rightarrow \sqrt{5x+9} + \sqrt{5x-9} = 6 + 3\sqrt{2}$$

$$\Rightarrow \sqrt{5x+9} + \sqrt{5x-9} = \sqrt{36} + \sqrt{18}$$

Comparing the L.H.S. and R.H.S.

$$\Rightarrow 5x + 9 = 36 \Rightarrow 5x = 27 \Rightarrow x = \frac{27}{5} \text{ (can be verified using the second term as well).}$$

$$\Rightarrow \sqrt{10x+9} = \sqrt{\left(10 \times \frac{27}{5}\right) + 9} = \sqrt{63} = 3\sqrt{7}$$

 VIDEO SOLUTION

43. D

It is given that $x^8 + \left(\frac{1}{x}\right)^8 = 47$, which can be written as:

$$\Rightarrow (x^4)^2 + \left(\frac{1}{x^4}\right)^2 = 47$$

$$\Rightarrow (x^4 + \frac{1}{x^4})^2 - 2 \cdot x^4 \cdot \frac{1}{x^4} = 47$$

$$\Rightarrow (x^4 + \frac{1}{x^4})^2 = 49$$

$$\Rightarrow x^4 + \frac{1}{x^4} = 7$$

Similarly, $x^4 + \frac{1}{x^4} = 7$ can be expressed as:

$$\Rightarrow (x^2)^2 + \left(\frac{1}{x^2}\right)^2 = 7$$

$$\Rightarrow (x^2 + \frac{1}{x^2})^2 - 2 \cdot x^2 \cdot \frac{1}{x^2} = 7$$

$$\Rightarrow (x^2 + \frac{1}{x^2})^2 = 9$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 3$$

By the same logic, we get $x + \frac{1}{x} = \sqrt{5}$

$$\text{Now, } x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right)$$

$$\Rightarrow x^3 + \frac{1}{x^3} = (\sqrt{5})^3 - 3\sqrt{5} = 2\sqrt{5}$$

By the same logic, we can say that

$$\Rightarrow x^9 + \frac{1}{x^9} = \left(x^3 + \frac{1}{x^3}\right)^3 - 3 \cdot x^3 \cdot \frac{1}{x^3} \left(x^3 + \frac{1}{x^3}\right)$$

$$\Rightarrow x^9 + \frac{1}{x^9} = (2\sqrt{5})^3 - 3(2\sqrt{5})$$

$$\Rightarrow x^9 + \frac{1}{x^9} = 40\sqrt{5} - 6\sqrt{5} = 34\sqrt{5}$$

The correct option is D

 VIDEO SOLUTION

44.7

It is given that $\log_{\sqrt{3}}(x) + \frac{\log_x(25)}{\log_x(0.008)} = \frac{16}{3}$, which can be written as:

$$\Rightarrow 2 \log_3 x + \log_{0.008} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{\frac{8}{1000}} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{\frac{1}{125}} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{5^{-3}} (5)^2 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x - \frac{2}{3} = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x = \frac{16}{3} + \frac{2}{3}$$

$$\Rightarrow 2 \log_3 x = 6$$

$$\Rightarrow \log_3 x^2 = 6 \Rightarrow x^2 = 3^6$$

$$\text{Hence, } \log_3 (3 \cdot x^2) = \log_3 (3 \cdot 3^6) = \log_3 3^7 = 7$$

 VIDEO SOLUTION

45.14

The first term of the expression can be rewritten as $\frac{\frac{1}{3} \log_2(a+b)}{\log_2 c}$

Using the property $\frac{m}{n} \log_a b = \log_a b^{\frac{m}{n}}$ this can be rewritten as

$$\frac{\log_2(a+b)^{\frac{1}{3}}}{\log_2 c}$$

And finally using the property $\frac{\log_b a}{\log_b c} = \log_c a$, we can rewrite the expression as

$$\log_c (a+b)^{\frac{1}{3}}$$

Doing identical operations in the second term, we get the entire left-hand side to be:

$$\log_c (a+b)^{\frac{1}{3}} + \log_c (a-b)^{\frac{1}{3}}$$

Using property $\log_c a + \log_c b = \log_c (ab)$ we get

$$\log_c \left[(a+b)^{\frac{1}{3}} (a-b)^{\frac{1}{3}} \right]$$

$$\log_c \left[(a+b)(a-b) \right]^{\frac{1}{3}}$$

$$\log_c \left[(a^2 - b^2) \right]^{\frac{1}{3}}$$

This expression is given to be equal to $2/3$

Using the definition of log: $\log_c N = a$ which is $c^a = N$

$$\text{we get: } c^{\frac{2}{3}} = (a^2 - b^2)^{\frac{1}{3}}$$

Cubing both sides:

$$c^2 = a^2 - b^2$$

$$\text{Finally giving } a^2 = b^2 + c^2$$

We have upper limits on b and c as 10, and we want to maximize the value of a squared.

This can be thought of as a right-angled triangle, and the value of a will be maximum when both b and c are equal to 10, giving $a^2 = 200$, but this would not give an integer value of a

We need to adjust a^2 to the biggest square less than 200, which is 196

Giving the value of a as 14.

Therefore, 14 is the correct answer.

 VIDEO SOLUTION

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46.11

Squaring on both sides, we get:

$$x + 6\sqrt{2} + x - 6\sqrt{2} - 2(x^2 - 72)^{\frac{1}{2}} = 8$$
$$x - (x^2 - 72)^{\frac{1}{2}} = 4$$

Bringing x to the other side, we get:

$$-(x^2 - 72)^{\frac{1}{2}} = 4 - x$$

Squaring on both sides again, we get:

$$x^2 - 72 = 16 + x^2 - 8x$$

$$8x = 88$$

$$x = 11$$

Therefore, 11 is the correct answer.

 VIDEO SOLUTION

47.31

Using the logarithmic property that $\log_{a^p} b = \frac{1}{p} \log_a b$

$$4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13$$

Can be written as

$$4 \log_{10} x + 2 \log_{10} x + \frac{8}{3} \log_{10} x = 13$$

$$\frac{26}{3} \log_{10} x = 13$$

$$\log_{10} x = 1.5$$

$$x = 10^{1.5}$$

$$x = \sqrt{1000}$$

$$[\sqrt{1000}] = 31$$

Where [.] is Greatest Integer Function since that is what is asked in the question,

31 is the greatest integer that does not exceed x.

 VIDEO SOLUTION

48.A

$(a + b\sqrt{n})$ is the positive square root of $(29 - 12\sqrt{5})$

$$\text{So } 29 - 12\sqrt{5} = (a + b\sqrt{n})^2$$

$$29 - 12\sqrt{5} = a^2 + b^2n + 2ab\sqrt{n}$$

$$a^2 + b^2n = 29 \text{ and}$$

$$ab\sqrt{n} = -6\sqrt{5}$$

$$a^2b^2n = 180$$

$$b^2n = \frac{180}{a^2}$$

Substituting this in the above equation,

$$a^2 + \frac{180}{a^2} = 29$$

$$a^4 - 29a^2 + 180 = 0$$

$$a^2 = \frac{(29 \pm \sqrt{29^2 - 4(180)})}{2}$$

$$a^2 = \frac{(29 \pm \sqrt{841 - 720})}{2}$$

$$a^2 = 9 \text{ or } 20$$

That means, one of a^2 or b^2n is 9 and 20.

We also have, $ab\sqrt{n} = -6\sqrt{5}$ that means one of a or b should be negative

And also the fact that this is a positive square root,

And we need to maximise the value of a, b and n.

We can have $a=-3$, $b=1$ and $n=20$.

This satisfies all the above equations, and the value of $a+b+n=18$.

 VIDEO SOLUTION

49. B

Opening the square on the left-hand side, we get $a^2 + 3b^2 + 2ab\sqrt{3}$

Comparing the rational part on both sides we get: $a^2 + 3b^2 = 52$

And comparing the irrational part we get: $2ab\sqrt{3} = 30\sqrt{3}$

$ab = 15$, Since we are given that a and b are natural numbers, the possible values of a and b are (1,15), (3, 5), (5, 3), or (15,1)

Putting these values in the first relation we got, we see that 15 squared would exceed the required value and would not be the case.

We need not check if $a=5$, $b=3$ or $a=3$, $b=5$ since the answer would be the same.

($a=5$ and $b=3$ would satisfy it)

$$a+b = 5+3 = 8$$

Therefore, Option B is the correct answer.

 VIDEO SOLUTION

50. A

Taking $10^x = a$

we get $a + \frac{4}{a} = \frac{81}{2}$

This would give the quadratic equation: $2a^2 - 81a + 8 = 0$

We want to find the sum of possible values of x, let the value of x be x_1 and x_2

these would correspond to $\log a_1$, and $\log a_2$

The sum of $\log a_1 + \log a_2$ would be $\log (a_1 \times a_2)$

From the quadratic equation we got above, we can see that the product of the possible values of a would-be $8/2 = 4$

Therefore, the sum of values of x would be $\log (4)$ which would be $2 \log_{10} 2$

Therefore, Option A is the correct answer.



51.2

Taking a log for each of the expressions, we get the following:

$$\log_3 4 = a, \log_4 5 = b, \log_5 6 = c, \log_6 7 = d, \log_7 8 = e, \log_8 9 = f$$

The expression $abcef$ would then be: $\log_3 4 \times \log_4 5 \times \log_5 6 \times \log_6 7 \times \log_7 8 \times \log_8 9$

Next, we can use this property of log: $\frac{\log_b a}{\log_b c} = \log_c a$

Using this, we get:

$$\frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times \frac{\log 6}{\log 5} \times \frac{\log 7}{\log 6} \times \frac{\log 8}{\log 7} \times \frac{\log 9}{\log 8}$$

All the terms will cancel out except: $\frac{\log 9}{\log 3} = \log_3 9 = 2$

Therefore, 2 is the correct answer.

52. D

For base of log in range $1/4 \in (0, 1)$ and $\log_{1/4}(x) > 0$ is true only if $0 < x < 1$.

For integer n , $x = n^2 - 7n + 11$ is an integer, so it cannot lie strictly between 0 and 1.

So, there is no integer value for which this inequality is satisfied.

53. D

Since the base of the logarithm has to be greater than zero and cannot be 1, x has to be greater than 3 and cannot be 4. Also, since $x^2 - 9 > 0$, x will be greater than 3.

The equation can be rewritten as;

$$\log_{x-3} (x^2 - 9) - \log_{x-3} (x + 1) = 2$$

Or,

$$\log_{x-3} \frac{x^2 - 9}{x + 1} = 2$$

$$\Rightarrow \frac{(x + 3)(x - 3)}{x + 1} = (x - 3)^2$$

$$\Rightarrow \frac{(x + 3)}{x + 1} = (x - 3)$$

$$\Rightarrow (x + 3) = (x - 3)(x + 1)$$

$$\Rightarrow x + 3 = x^2 - 3x + x - 3$$

$$\Rightarrow x^2 - 3x - 6 = 0$$

The roots of the quadratic above are $\frac{3 \pm \sqrt{3^2 + 24}}{2}$

The negative value of x will not be possible, the positive value that satisfies is $\frac{3 + \sqrt{33}}{2}$

The correct answer is option D.

 VIEW SOLUTION

54. **A**

$$64 = 8^2 \text{ and } 512 = 8^3$$

$$\log_{64} x^2 + \log_8 \sqrt{y} + 3 \log_{512} (\sqrt{y}z) = 4,$$

$$\text{Property of log: } \log_{b^m} a^n = \frac{n}{m} \log_b a$$

$$\log_{8^2} x^2 + \log_8 \sqrt{y} + 3 \log_{8^3} (\sqrt{y}z) = 4$$

$$\text{Using the above-mentioned property, the expression becomes } \log_8 x + \log_8 \sqrt{y} + \log_8 (\sqrt{y}z) = 4$$

$$\log_8 x \sqrt{y} \cdot (\sqrt{y}z) = 4$$

$$\log_8 xyz = 4$$

$$xyz = 8^4 = 2^{12}$$

Using AM-GM inequality

$$\frac{(x+y+z)}{3} \geq \sqrt[3]{xyz}$$

$$\frac{(x+y+z)}{3} \geq 2^4$$

$$(x + y + z) \geq 48$$

 VIDEO SOLUTION